March 20, 2026

The Editor
Classical and Quantum Gravity

Dear Editor,

I am submitting the enclosed letter, "The Coordinate-Space Geometry of the Tetrad Frame," for consideration in *Classical and Quantum Gravity*.

The tetrad has been the standard variable for coupling gravity to fermionic matter since the work of Cartan, Einstein, and Weyl. It is central to Poincaré gauge gravity, Ashtekar variables, loop quantum gravity, bimetric gravity, and supergravity. In over a century of use, the spatial geometry generated by its local frame has not, to our knowledge, been explicitly derived in coordinate space.

This submission identifies a previously unreported coordinate-space geometry associated with the local tetrad frame. Starting from a single orthonormal tetrad, the paper shows that the induced half-space structure yields exactly two bounded tetrahedral regions whose union forms a stellated octahedron with a regular octahedral core. The opposite-oriented tetrahedron is not introduced independently; it emerges as the unique second bounded region of the same construction. Because the result is kinematic and tetrad-level, it arises prior to any choice of gravitational action or field equation and is therefore relevant across multiple tetrad-based formalisms.

The geometry is established by two complementary derivations: a convex boundedness classification in the main letter, and a supplementary group-theoretic derivation from the quadratic tetrad-to-metric relation. Supporting materials also include a supplementary note deriving the tetrahedral frame from the Lorentz signature's isotropy, an interactive rendering, and a validation script confirming the principal geometric claims.

I believe referees with expertise in tetrad formalism, vierbein geometry, or Poincaré gauge gravity would be well placed to evaluate this work.

Yours sincerely,

Stephen Nelson
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